

Exploiting the Overnight Market Anomaly in Indian Markets: Can Excess Returns Survive Market Frictions

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ABSTRACT

This study examines the practical exploitability of the overnight market anomaly in Indian equity markets using index futures. While prior research [4][12] suggests that most equity returns are generated outside trading hours, the economic viability remains uncertain once transaction costs, slippage, and taxes are accounted for. Using the past six-year historical data of the NIFTY future contracts to backtest a systematic trading strategy that maintains long exposure exclusively during overnight periods, entering positions near the market close and exiting them at the subsequent market open. The analysis incorporates realistic assumptions surrounding leverage, margin requirements, transaction costs [9], slippage, and statutory charges in the Indian derivatives market. The performance of the strategy is evaluated relative to a passive buy-and-hold exposure to the NIFTY Index and assessed using cumulative returns, volatility, Sharpe ratio [11], maximum drawdown [8], and risk-adjusted performance measures. The result provides evidence on the extent to which overnight returns contribute to the Indian equity market and whether this strategy can survive realistic market friction. The study further describes the practical challenges in converting a documented anomaly into an investment strategy

Keywords: Overnight Return Anomaly, Indian Equity Markets, Return Seasonality, Trading Frictions and Transaction Costs, Market Efficiency, Systematic Trading Strategies, Risk-Adjusted Performance Measurement

1. INTRODUCTION

The efficient market theory [1][2] suggests that asset prices already reflect all available information at all times, making it impossible for any individual or institution to always beat the market, but this theory doesn't always hold because of some persistent anomalies [3] that challenge market efficiency, one of which is the irrationality and herd mentality among retail participants of the market [6] that often deviates the market from its fundamental value. The anomaly we will be focusing on here is the overnight return anomaly [4], which is the most persistent and widely documented of all.

Evidence from several developed and emerging markets [12] [13] suggests that a disproportionately large share of long-term equity returns is generated during non-trading hours, while intraday returns contribute relatively little to overall returns. This finding raises an important question regarding the degree of market efficiency. The existing explanation proposes that the existence of this anomaly primarily comes from information releases [5] such as earnings announcements, macroeconomic events, and global market development that occur frequently beyond trading hours. In addition to this, order imbalances and unexecuted trades that persist beyond trading hours artificially inflate the price when the market reopens, potentially contributing to overnight price adjustments.

While the overnight anomaly has garnered considerable attention in the international market, relatively few studies have been made on its exploitability within the Indian market. More importantly, much of the existing literature focuses on identifying the anomaly rather than evaluating whether it can be transformed into a profitable trading strategy after accounting for real-world implementation costs [9].

A strategy that may seem statistically significant may prove to be unprofitable once brokerage charges, taxes, slippage, and other market frictions are considered. The primary objective of this study is to determine whether the excess return associated with overnight price movements remains economically meaningful after accounting for implementation costs [9]. By focusing on the index futures markets, rather than solely focusing on the return pattern, this paper contributes to the literature on market anomalies by bridging the gap between statistical evidence and practical investment application.

2. LITERATURE REVIEW

The Efficient Market Hypothesis states that asset prices fully reflect all available information, making it near impossible for any individual to generate abnormal risk-adjusted returns [2]. Still, decades of empirical research have documented persistent anomalies [3] that are inconsistent with the above hypothesis. Such anomalies have attracted considerable academic attention because they challenge traditional pricing theories, potentially paving the way for opportunities to develop systematic trading strategies.

Traditional studies evaluate daily returns without distinguishing between trading and non-trading hours. More recent research decomposes total daily returns into overnight and intraday components [10]:

$$R(\text{daily}) = R(\text{intraday}) + R_{\text{ov}} (\text{overnight})$$

Overnight return is defined as:

$$R_{\text{ov}} = \ln(O_t / C_{t-1})$$

Intraday return is defined as:

$$R_{\text{in}} = \ln(C_t / O_t)$$

Total daily return is:

$$R = \ln(C_t / C_{t-1})$$

Here, (O_t) and (C_t) represent the opening and closing prices on day (t), respectively

Multiple bodies of academic literature have documented the existence of systematic differences between overnight and intraday returns on US equities and ETFs [7]; however, the extent to which such patterns persist across different market structures and geographical regions remains an active area of research. To examine whether similar return dynamics are present in the Indian market, this study analyzes the performance of the NIFTY 50 Index by segregating total returns into overnight and intraday components. The analysis seeks to determine whether overnight returns account for a disproportionate share of overall market performance and whether such returns can be practically captured through a futures-based trading strategy

Table 1. Overnight and Intraday Log Returns of the NIFTY 50

Total Return	Intraday Return	Overnight Return
84.37%	-71.96%	156.33%

Using daily OHCLV NIFTY 50 data from 20 June 2020 to 21 June 2026, it was noticed that NIFTY 50 generated a cumulative return of 84.37% over the sample period, 156.33% of this return was attributable to overnight price movements, whereas intraday returns contributed -71.96%.

These findings provide preliminary evidence of an overnight market anomaly [12] in the Indian equity market, thus motivating further study of whether this can be economically exploited through index futures after accounting for transaction costs and market friction. Despite the extensive documentation of the overnight return anomaly, there is no universally accepted reason for its persistence. Existing literature suggests that this phenomenon is a combination of information asymmetry and behavioural bias. These reasons aren't mutually exclusive and could contribute together to the observed return during non-trading hours.

One of the most accepted reasons is the information arrival hypothesis [5], since markets operate within fixed hours. In contrast, information relevant to the asset prices, like earnings announcements and geopolitical developments, often takes place outside trading hours. Consequently, prices adjust to new information at the subsequent market open [14], leading to concentration of returns during the overnight period. Behavioural bias [6] also offers a potential explanation for this anomaly. Institutions often execute large orders before the market closes to rebalance portfolios, and retail participation is disproportionately high at market open as a reaction to the overnight information. Such systematic trading patterns can generate temporary price pressure, contributing to persistent differences between overnight and intraday returns.

Although each of these explanations provides really detailed insight, no single theory completely accounts for the magnitude and persistence of the overnight return anomaly across different markets and time periods.

While there exists a considerable amount of literature documenting the existence of this anomaly across equity markets [4][12][13], comparatively fewer studies examine whether this anomaly remains economically exploitable after accounting for realistic trading costs. A strategy may generate positive gross returns but could be unprofitable once implementation costs are considered. This study focuses on constructing and backtesting a systematic overnight strategy using NIFTY Index Futures from June 2020 to June 2026. Unlike prior studies focusing on gross return differentials, this research incorporates realistic transaction costs, statutory charges, and execution frictions to evaluate the economic viability of exploiting the overnight anomaly. The performance of this strategy is assessed relative to a buy-and-hold strategy benchmark using both absolute and risk-adjusted performance measures.

3. METHODOLOGY

Having established the existence of this anomaly in the Indian equity market, the subsequent analysis examines whether this can be transformed into a profitable and practically implementable investment strategy. A systematic overnight trading strategy is constructed using NIFTY Index futures, providing liquid and efficient exposure to the underlying index.

This section outlines the framework employed to implement and execute this strategy. Covering the trading rules, portfolio construction methodology, transaction cost assumptions, and performance evaluation metrics presented in detail. The performance of the proposed strategy is compared to a passive buy-and-hold benchmark using both absolute and risk-adjusted performance measures

3.1. Data Collection

This study utilises daily OHCLV data of the NIFTY50 Index and NIFTY Index Futures, covering the period from 20 June 2020 to 21 June 2026, obtained from the website of the National Stock Exchange (NSE) of India. The sample period considered for this study spans a variety of market conditions, including the post-pandemic recovery, periods of elevated inflation and market correction, thereby providing a complete environment to test the robustness of this strategy. Furthermore, this study will use the nearest to maturity NIFTY Index Futures contract due to its liquidity and bid-ask spreads, and the positions are rolled over to the subsequent contract prior to expiry

3.2. Sample Filtering

To ensure that the results are not driven by extraordinary market conditions or institutional features unrelated to the overnight anomaly, certain trading days referred to as event days are excluded from the analysis. Major events like the general election result and other macroeconomic events that generate unusual price movements are omitted. Thursdays are also omitted from our analysis, due to elevated volatility and artificial price movements as a result of option expiry-related positioning and contract settlement effects.

3.3. Capital Allocation & Position Sizing

Both the strategy and passive buy-and-hold benchmark have a starting capital of ₹100,000 to ensure easy comparability of performance. For the overnight strategy, only 50% of the capital is deployed into each trade, introducing a conservative leverage assumption. Given below is the equation to find the maximum number of future units that can be acquired after taking into consideration margin constraints. Assuming margin requirements for a futures contract is 15%

Position Sizing for Overnight Strategy

$$(\text{Maximum Future Units}) N_t = \text{floor}(0.5 \cdot V_{t-1} / 0.15 \cdot F_{t-1}^{\text{close}})$$

Where V_{t-1} denotes the portfolio value at the end of period $t-1$, and F_{t-1}^{close} is the futures closing price.

The buy-and-hold strategy is constructed by investing the initial capital of ₹100,000 in the NIFTY 50 at the beginning of the sample period and holding the same position throughout the investment horizon without rebalancing. Given below is the equation to determine the maximum number of units purchased.

Position Sizing for Buy-and-Hold Strategy

$$(\text{Maximum Units}) N_{bh} = \text{floor}(100000 / P_0)$$

Where P_0 denotes the initial value of the NIFTY50.

This framework ensures that both strategies start with identical capital while allowing the overnight strategy to compound returns through periodic reinvestment.

3.4. Strategy Design

To evaluate the practical exploitability of the documented overnight return anomaly, a systematic strategy is designed using NIFTY Index futures to obtain exposure exclusively during overnight periods while remaining uninvested during regular trading hours

The overnight strategy follows a simple rule-based framework:

1. At the close of trading on day $(t-1)$, a long position is established in the nearest-to-maturity NIFTY Index Futures contract
2. The position is maintained throughout the overnight period, thereby capturing the close-to-open price movement
3. At the market open on day (t) , the futures position is liquidated, and the realized profit or loss is added to the portfolio value.
4. The process is repeated on each trading day other than event days over the sample period, with profits and losses being fully reinvested, resulting in a compounded return series

3.5. Transaction Cost Framework

The profitability of the strategy can be significantly affected by transaction costs and market frictions. Since the proposed strategy involves entering and exiting positions daily, even minute execution costs can erode cumulative returns over the investment horizon [9].

The total transaction cost incurred over trade (t) is defined as:

$$TC(t) = BC(t) + STT(t) + ETC(t) + SD(t) + GST(t) + SEBI(t)$$

Where TC denotes transaction cost, BC denotes brokerage charge, STT denotes Securities Transaction Tax, SD denotes stamp duty, ETC denotes exchange transaction charges, and SEBI denotes regulatory charges per trade executed.

Table 2. Transaction Cost Breakdown

BC	STT	ETC	SD	GST	SEBI
20/ trade	0.01%	0.0018%	0.002%	18%	0.0001%

All percentage-based charges are applied to the notional value of the futures position, except GST, which is levied on the sum of brokerage and exchange transaction charges

3.6. Performance Evaluation

The performance of the overnight trading strategy is evaluated relative to a passive buy-and-hold investment in the NIFTY 50 Index using both absolute and risk-adjusted performance measures

The cumulative return of each strategy is computed as:

$$CR = (V_t / V_0) - 1$$

Where V_0 denotes the initial portfolio value, and V_t denotes the terminal portfolio value.

Now, to facilitate comparison on an annual basis, the compounded annual growth rate (CAGR) is calculated as

$$CAGR = (V_t / V_0)^{(1/n)} - 1$$

Where V_0 and V_t denote the same as above, and n is the investment horizon in years

Portfolio risk is measured using its annualised volatility, i.e., the annualised standard deviation of the returns

$$\text{Portfolio Risk } (\sigma_p) = \sigma(\text{daily}) * \sqrt{252}$$

Where $\sigma(\text{daily})$ denotes the standard deviation of daily returns, and 252 represents the average number of trading days in a year.

The Sharpe Ratio measures the excess return generated per unit of total risk undertaken by the strategy [11]. Risk-adjusted performance is examined using the Sharpe ratio given by:

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

Where R_p denotes annualized portfolio return, R_f denotes the risk-free rate (10y Government Bond Yield), and σ_p denotes the annual volatility of the portfolio

Maximum drawdown indicates the largest cumulative loss an investor would have experienced in a strategy [8]. To evaluate this downside risk, the Maximum Drawdown (MDD) is computed as:

$$\text{MDD} = \max((\text{Peak} - \text{Trough}) / \text{Peak})$$

Where Peak and Trough represent the highest and lowest portfolio values during the sample period.

4. RESULT AND DISCUSSION

Having established the empirical existence of the overnight market anomaly and the construction of a systematic futures-based trading strategy to capture it, this section presents the results of the backtest, evaluating the profitability, risk, and economic significance of the strategy while comparing it with the performance of a passive investment in the NIFTY 50 Index

4.1. Evidence of the Overnight Market Anomaly

Table 1 demonstrates that the NIFTY 50 generated a cumulative logarithmic return of 84.37% during the sample period. Remarkably, cumulative overnight logarithmic returns accounted for 156.33% of this total performance while intraday returns contributed -71.96%. These findings suggest that most of the wealth creation takes place during the non-trading hours [7], providing strong evidence for the existence of a persistent overnight market anomaly.

4.2. Performance of the Strategy

Figure 1. Comparative Performance of Both Strategies



Table 3. Performance Comparison of the Overnight Trading Strategy and the Buy-and-Hold Benchmark

Metric	Overnight Strategy	Buy & Hold
Initial Capital	₹100,000	₹100,000

Final Portfolio Value	₹427,980.72	₹233,472.34
Cumulative Return	328%	133.5%
CAGR	27.47%	15.09%
Annualized Volatility	19.54%	15.47%
Sharpe Ratio	1.01	0.56
Maximum Drawdown	-16.02%	-17.90%

The result indicates that the systematic exploitation of overnight price movement produced superior absolute returns over the sample period.

The performance is further reflected in its annualised return characteristics. The overnight strategy achieved a CAGR of 27.47%, compared to 15.09% for the benchmark portfolio. Although the strategy exhibited higher annualized volatility (19.54% versus 15.47%), it generated considerably greater returns per unit of risk, as evidenced by its Sharpe ratio of 1.01. From a downside risk perspective, the strategy demonstrated good resiliency. The maximum drawdown experienced by the overnight strategy was -16.02%, marginally lower than the -17.90% observed for the benchmark portfolio. Additionally, the strategy exhibited improved capital preservation characteristics during periods of market stress.

4.3. Impact of Transaction Costs

Table 4. Performance Comparison of the Overnight Trading Strategy and the Buy-and-Hold Benchmark

Metric	Gross Strategy	Net Strategy
Cumulative Return	746.6%	328%
CAGR	42.84%	27.47%
Sharpe Ratio	1.59	1.01

While transaction costs reduced the profitability of the strategy, excess returns remained economically significant even after accounting for brokerage charges, taxes, exchange fees, and slippage. This suggests that the overnight market anomaly remains an exploitable market inefficiency.

4.4. Discussion and Interpretation

The results are consistent with international literature documenting the dominance of overnight returns in equity markets. However, unlike previous studies that focused only on the statistical significance [12], the present study demonstrates how this anomaly could be exploited economically by a futures-based implementation, which remains profitable after incorporating market frictions. These findings challenge the market efficiency and suggest that information is incorporated into prices primarily during non-trading periods.

5. LIMITATIONS

Despite the encouraging findings, the study is prone to several limitations. Firstly, the analysis is conducted over a limited sample period from June 2020 to June 2026, which may not capture the behaviour of the market across different economic and market cycles. The study further assumes a constant margin requirement of 15% and fixed cost parameters, which may vary depending on market conditions and regulatory changes. The study also assumes execution at the official opening and closing prices of the futures contracts. Actual implementation may be affected by execution delays and liquidity constraints. The analysis focuses exclusively on the NIFTY50 Index and its future contracts; the results may not necessarily generalise to other asset classes or international equity markets.

6. CONCLUSION

This study examined the existence and practical exploitability of the overnight return anomaly [12] in the Indian equity market. Analysis revealed that a significantly large share of cumulative market returns was generated during non-trading hours, while intraday returns contributed negatively over the sample period.

Based on this evidence, a systematic trading strategy was implemented using NIFTY Index Futures to capture overnight price movements. The results demonstrate that the proposed strategy substantially outperformed a passive buy-and-hold investment by 194.5%. Additionally, the strategy demonstrated superior risk-adjusted returns. Importantly, these results remain profitable after incorporating transaction charges and market frictions, suggesting that this anomaly is not merely a statistical irregularity but an economically exploitable market inefficiency.

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